

31. Mouse Events (Basics)

Ye events sirf mouse se nahi, phones/tablets par bhi (compatibility ke liye) emulate hote hain.

Mouse event types

- `mousedown / mouseup`: button dabaya / chhoda
- `mouseover / mouseout`: pointer element par aaya / gaya
- `mousemove`: har move par
- `click`: `mousedown` + `mouseup` **same element par** (sirf left button)
- `dblclick`: do baar jaldi click (ab kam use)
- `contextmenu`: right click (keyboard ke special key se bhi aa sakta hai)

Events ka order (fixed)

Left click: `mousedown` -> `mouseup` -> `click`

Mouse button: `event.button`

`mousedown/mouseup` me kaam aata hai (ye kisi bhi button par chalte hain).

Button	<code>event.button</code>
Left	0
Middle	1
Right	2
X1 (back)	3
X2 (forward)	4

- `event.buttons` (ek saath dabe hue buttons) rarely use hota hai.
- `event.which` **purana/deprecated** hai, use mat karo.

Modifier keys

Har mouse event me ye properties hoti hain (pressed ho to `true`):

- `shiftKey`, `altKey` (Mac par Opt), `ctrlKey`, `metaKey` (Mac par Cmd)

```
js
```

```
button.onclick = function(event) {  
  if (event.altKey && event.shiftKey) alert('Hooray!');  
};
```

Mac ka dhyan: Windows/Linux me `Ctrl` jo kaam karta hai, Mac me `Cmd`. Mac par `Ctrl`+click actually **right click** (`contextmenu`) ban jaata hai. Isliye dono check karo:

```
js  
  
if (event.ctrlKey || event.metaKey) { ... }
```

Mobile par keyboard nahi hota, isliye modifier keys sirf **extra** feature ke roop me rakho.

Coordinates

- `clientX / clientY`: window ke hisaab se (`position:fixed`) jaisa)
- `pageX / pageY`: document ke hisaab se (scroll se nahi badalte)

Selection rokna

Double click ya mouse dabakar ghumane se text **select** ho jaata hai (kabhi kabhi unwanted). Rokne ke liye `mousedown` par default action roko:

```
html  
  
<b onclick="alert('Click!')" onmousedown="return false">Double-click me</b>
```

(Text ke andar se select shuru nahi hoga, par pehle/baad se ho sakta hai.)

Copy rokna ho to `oncopy` par `return false`. (Par user page source se phir bhi le sakta hai.)

Summary

- Button: `button`
- Modifiers: `altKey`, `ctrlKey`, `shiftKey`, `metaKey` (Mac ke liye `metaKey || ctrlKey`)
- Coordinates: `clientX/Y` (window), `pageX/Y` (document)
- `mousedown` ka default action = text selection, zarurat ho to roko.

Task

Selectable list: click par sirf wahi `.selected`; `Ctrl`/`Cmd` ke saath click par sirf us item ko toggle karo. Native text selection `mousedown` par roko.